

MetroPredict

Better replenishment suggestions. A measured case for a selective pilot.

Prepared for Eshwar Vudhanthi | SQL + Python + Tableau | September 2026

Evidence	Measured result
Public transaction coverage	1,469,307 rows 2,469 households
Held-out ranking gain	8.5% relative improvement over frequent purchases
Fixed list accuracy	16.47% Precision@8 1.32 matches per shopping day
Short-list policy	36.4% item precision 57% eligible-day coverage

Recommendation: conditional, limited pilot; no full launch. The model clears the project's relative-accuracy gate, but an eight-item list is too noisy to treat as a complete shopping list. A validation-selected confidence threshold offers a more credible customer experience.

Commercial finding: recommendation acceptance is not incremental sales. The central illustrative economics are negative, including under the shorter-list policy. Validate convenience, adoption, net purchase change and cost before investment.

The study is inspired by Metro Market purchase history. It uses public Complete Journey data and has no retailer affiliation. Personal Metro Market receipts are not included; the import workflow is ready for a later extension.

Data you can audit

The analyzed edition is the completejourney R package: a one-year slice of the original study. Its preparation script maps dates onto 2017 and modifies some metadata and discount fields. Those dates provide a relative timeline, not current trading evidence. No holiday or day-of-week features are used.

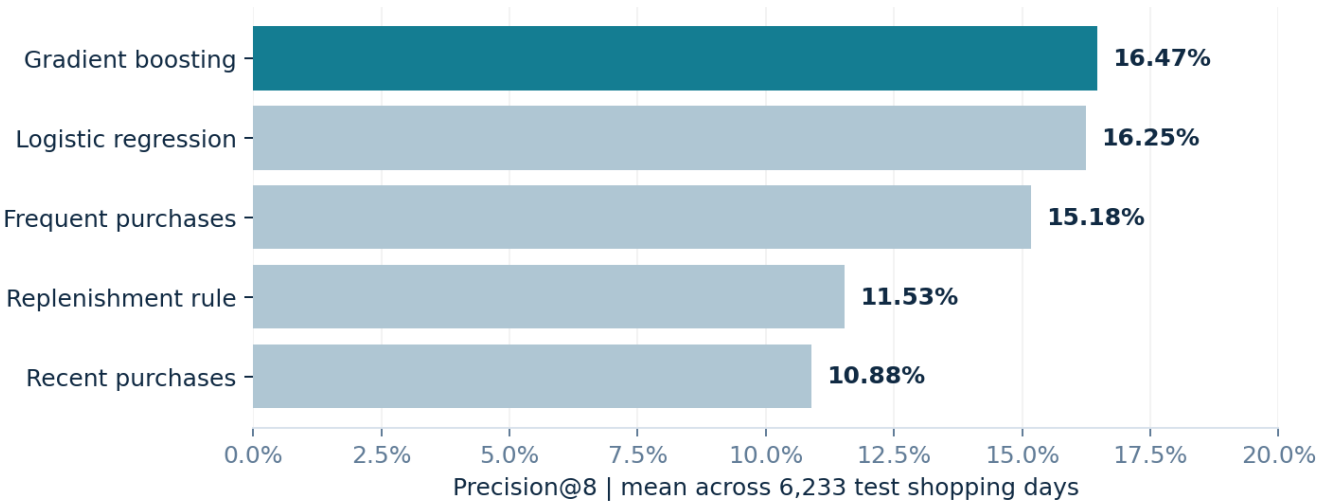
Source / scope	Audit result
Transactions	1,469,307 receipt lines; 155,848 baskets; \$4,596,039.58 recorded sales
Household shopping days	127,347 across valid sales rows; multiple same-day receipts combined
Excluded purchase rows	8,869 rows with quantity ≤ 0 or negative sales
Product metadata	92,331 products; 4,836 transaction rows lack product metadata
Coupon mapping	116,204 rows including 4,872 exact duplicates; do not expand sales with an unqualified coupon join
Campaign data	27 campaigns; 19 fully within the observation window; 8 boundary campaigns

Modeling scope: grocery-related departments with known categories. A deterministic hash sample selects 800 households observed before April. The full data remains in the SQL analysis. Test results describe returning households in this sample, not cold-start customers or every retailer customer.

Time window	Use
January-March	History initialization and sampling pool
April-September	Training: every third eligible shopping occasion
October	Validation: model and baseline selection
November-December	Fixed-model test; histories update only after each observed day

SQLite warehouse: transactions, products, campaigns, campaign_descriptions, coupons and coupon_redemptions. SQL views cover clean sales lines, household days, customer summaries and purchase intervals. Full-period interval summaries are descriptive; prediction features are recalculated from past history only.

The model beats frequency, modestly



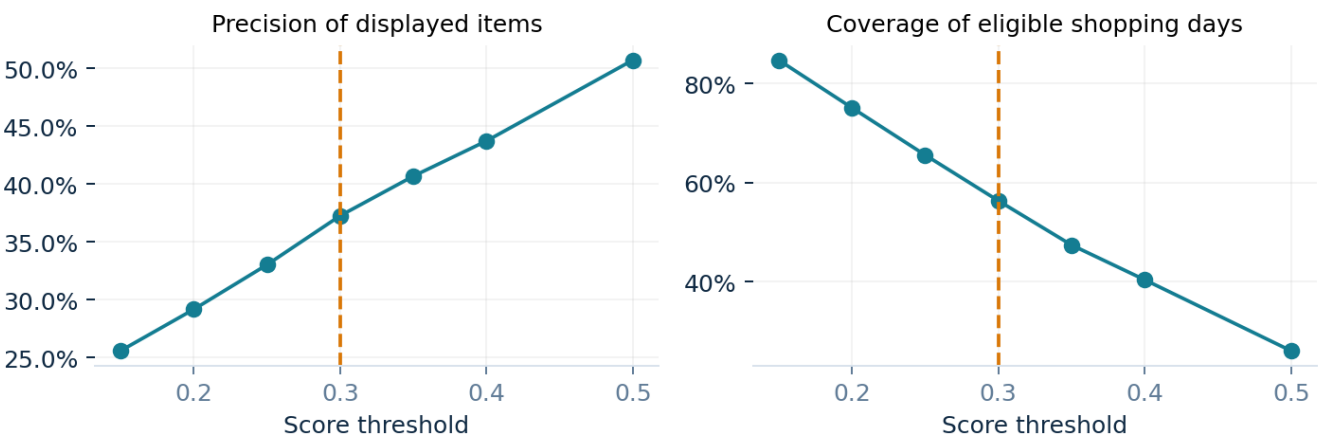
Test approach	Precision@8	Full-basket recall
Frequent purchases	15.18%	14.32%
Recent purchases	10.88%	10.41%
Replenishment rule	11.53%	10.69%
Logistic regression	16.25%	15.21%
Gradient boosting	16.47%	15.34%

Absolute gain: 1.30 percentage points, equivalent to about 0.104 additional matched items per shopping day. The household-clustered bootstrap 95% interval is 1.06-1.54 points. This is prediction gain, not a causal business effect.

The predeclared project gate was at least 5% relative Precision@8 improvement over the validation-selected simple baseline, with a clustered interval above zero. Gradient boosting clears that gate. Logistic regression is close enough to remain a lower-complexity deployment candidate.

Each eligible day has at least five prior grocery shopping days and eight candidates. Up to 120 previously purchased products from the last 180 days are ranked. Exact product ID defines a hit. New items and omitted candidates are included in full-basket recall, so its denominator remains demanding.

Show fewer suggestions when uncertain



Exploratory policy: after the primary benchmark, thresholds were compared on validation data. Select the threshold with maximum coverage among those reaching 35% displayed-item precision and at least 100 covered validation days. The selected score threshold was 0.30.

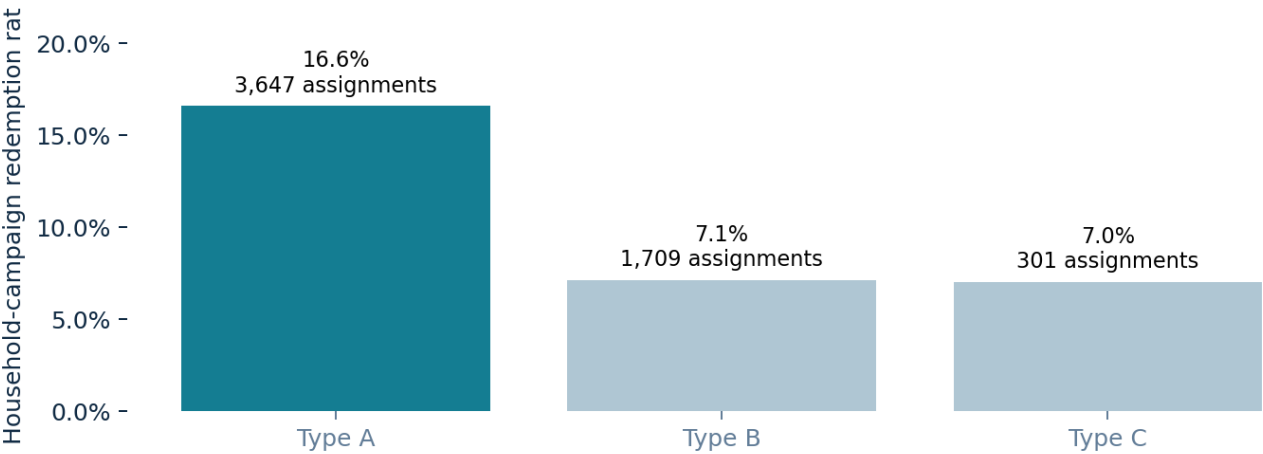
Fixed-policy test result	Value
Correct displayed suggestions	36.4% of 10,168 suggestions
Coverage	3,553 of 6,233 eligible days (57.0%)
List length on covered days	2.86 suggestions on average

These metrics differ from Precision@8: displayed-item precision weights suggestions, and coverage declines when the system abstains. Even the filtered list is wrong about 64% of displayed items. Present it as optional reminders, not a list of known household needs.

Category finding: among the most frequently suggested categories, tropical fruit reaches 27.1% precision versus 21.8% for fluid milk and 15.9% for eggs under the fixed-eight policy. These are descriptive segments; they do not prove category-specific lift.

Larger realized baskets have higher Precision@8, but their size is known only after shopping. This segment is a diagnostic, not a deployable targeting rule. A later version would need a forecast of basket size based on earlier trips.

Coupon response is evidence for testing



Across fully observed campaigns, Type A has a 16.6% household-campaign redemption rate, compared with 7.1% for Type B and 7.0% for Type C. The denominator is assigned households for each campaign, not redeemed coupon-product rows.

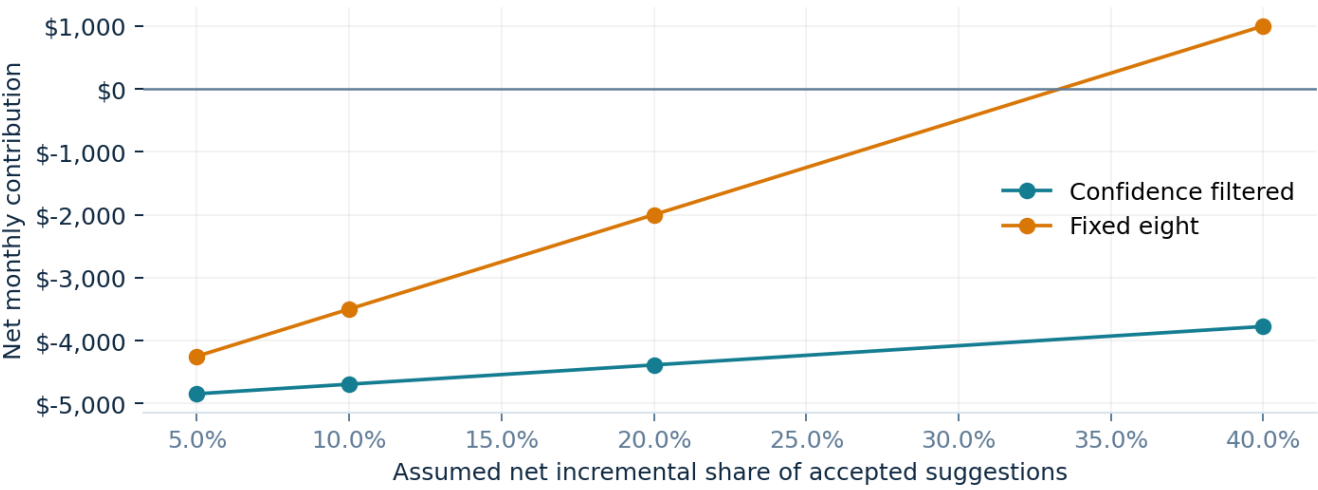
What this establishes: response differs across observed campaign groups. **What it does not establish:** that Type A caused more incremental purchases, is more profitable, or should receive more budget. Campaign types may differ in recipients, offers, duration and coupon availability.

Guard against	Treatment in this study
Boundary bias	Exclude 8 boundary campaigns from the type comparison
Join multiplication	Aggregate assignments and redemptions before joining; retain raw coupon mapping duplicates in audit
Exposure confusion	Assignment records are not evidence a shopper opened or saw an offer
Cost confusion	Separate retailer loyalty discounts, manufacturer coupons and retailer coupon matches

A separate SQL output compares each recipient's 28-day pre-campaign spending with the first 28 campaign days for eligible campaigns. This is a descriptive change only: time trends, overlapping campaigns and targeting remain unresolved. No causal uplift model is claimed.

Data checks found zero redemption records without a campaign assignment and zero redemptions outside the corresponding campaign dates. These checks validate joins; they do not establish random assignment.

Economics: the central case is negative



Illustrative input	Central assumption
Eligible shopping days / month	100,000
Feature adoption / item acceptance	25% / 10%
Net incremental share of accepted items	10%, after substitution and timing effects
Item value / contribution margin	\$3.00 / 25%
Monthly operating + allocated build cost	\$2,500 + \$30,000 / 12 = \$5,000

Fixed eight: \$-3,500 monthly net contribution. Break-even requires 33.3% of accepted suggestions to be incremental under these assumptions.

Confidence-filtered capacity: 1.63 suggestions per eligible day, including abstentions. Central net contribution is \$-4,694; break-even incremental share is 163%, which is infeasible. Lower cost, higher adoption or a different source of value would be required.

The 432 scenarios vary adoption, acceptance, incrementality, margin, coupon cost and list capacity. These are scenarios, not forecasts. Retention value is excluded. Manufacturer coupon amounts are not automatically charged to retailer profit. Coupon scenarios charge the illustrative retailer-funded discount to every accepted suggestion.

A retailer-facing pilot proposal

Decision owner: grocery digital product or loyalty team, with finance and customer research. The current result supports a small, budget-capped learning pilot if the team values the convenience hypothesis. It does not support a revenue-led rollout at the central assumed cost.

Household-randomized arm	Purpose
A: existing purchase-history experience	Establish business-as-usual behavior
B: confidence-filtered reminders	Measure reminder value without discounts
C: same reminders + defined coupon policy	Measure the additional effect and cost of offers

Primary commercial outcome: contribution per assigned eligible household over a prespecified follow-up horizon. Analyze by intention to treat, including people who never open the feature. Keep each household in one arm, stratify on prior shopping frequency, and use household-level uncertainty estimates.

Customer outcomes: list-building time, perceived usefulness, dismissals and opt-outs. Record suggestion exposure, acceptance and purchases separately. Track substitutions and later purchases to detect stock-up or pull-forward effects.

Before starting: use retailer baseline variance and a business-defined minimum useful effect to calculate sample size. Set duration to cover multiple normal purchase cycles, with a post-offer follow-up. Define cost, margin and customer-experience stopping rules before looking at results.

Launch only if: measured customer benefit and contribution meet the agreed threshold, service cost fits the budget, and important customer groups are not harmed. Modify the policy or stop if those conditions are not met. This experiment is a design deliverable; no live test has been run.

Practical edge: purchase frequency remains the strongest signal, while timing adds value. The product opportunity is choosing when to show a small set of optional reminders and when to abstain.

Reproduce, present, and extend

Deliverable	Location in project
Start here and measured results	README.md
Executable SQL and database	sql/ and data/processed/metropredict.sqlite
Python pipelines and saved models	src/ and models/
Executed analysis notebook	notebooks/MetroPredict_Analysis.ipynb
Editable Tableau workbook	tableau/MetroPredict.twbx
Metrics, campaign tables and scenarios	outputs/
Your Metro Market data later	personal/purchases_template.csv and src/personal.py

Validation status: source totals reconcile; historical features precede their targets; ranking and financial boundary tests pass. The Tableau workbook passes the official 2026.1 local-element XML schema with minimal declarations for unused imported namespaces. Native Tableau Desktop/Cloud opening and rendering were unavailable and remain unverified.

Interview framing: built an auditable retail pipeline across 1.47 million transactions, compared three simple rules with two machine-learning approaches, and found an 8.5% relative ranking improvement. Proposed a confidence-filtered pilot while explicitly separating predictions, observed coupon response and hypothetical economics.

Personal extension: import purchase dates, product names, quantities and sales values. Receipt and product IDs are optional. The default uses transparent personal-history rules; applying the public-data model is an explicitly unvalidated transfer. No personal receipts are included in this release.

Sources and documentation

[Complete Journey package and data](#)

[Source preparation and normalized calendar](#)

[Tableau official workbook schemas](#)

[Tableau packaged workbooks](#)

Data: Complete Journey, originating from 84.51 degrees, distributed by Brad Boehmke and Steven Mortimer through completejourney (package declares CC0). Raw checksums and source provenance are included. No Metro Market performance claim or affiliation is implied.